

Outline

Pseudo-Random Number Generators

Generating Uniform and Non-Uniform Random Variables

Inverse Transform Method

Acceptance-Rejection Method

Box-Muller Transformation

Derivation

Basic Algorithm



Ratio-of-Uniforms Method

Diagnosing Randomness

Simulation Error

Unit Summary

$\Rightarrow$ Recall:

goal: Trying to simulate random number from
a distⁿ of our interest. $F \rightarrow X \sim F$
 $\rightarrow f$

$X_1, \dots, X_n \sim F, f$
 $\downarrow$ CDF $\rightarrow$ PDF/pmf

$\hookrightarrow$ LCG:
modified $\rightarrow U_1, \dots, U_n \sim U[0,1]$ kmv
 $\downarrow$
 $X_1, \dots, X_n \sim F \rightarrow$ distⁿ of interest

1) Inverse transformation method.

$\therefore X = F^{-1}(U), \quad U \sim U[0,1]$

$\rightarrow$ CDF of X is F $\rightarrow$ PIT

AI $\Rightarrow U_1, \dots, U_n \sim U[0,1]$

$X_1 = F^{-1}(U_1), \dots, X_n = F^{-1}(U_n)$

$\rightarrow$ Sample from F

Pros: very Simple Alg.

Cons: inverse does not exist $\times$
(?)

(11) Acceptance Rejection method

$\Rightarrow$ goal: simulate from f $\Rightarrow \frac{f(y)}{c g(y)}$
 target/proposal $\rightarrow g \rightarrow$ you know how to deal.

e.g. f : Normal
 g : Cauchy

choice of g :

- 1) subset of g : same as f
- 2) choice of $c \rightarrow c = \sup_x \frac{f(x)}{g(x)}$

 $\hookrightarrow$ Acc rate = $\frac{1}{c}$

Alg
 $\Rightarrow$

- 1) $U \sim U[0,1]$
- 2) $Y \sim g$

(3) if $U \leq \frac{f(Y)}{c g(Y)} \rightarrow \text{Then } X = Y$
 o.w. go to step 1

Why Normal Variables Need a Special Trick

- The Normal CDF $\Phi(x)$ has **no closed-form inverse** $\Rightarrow$ inverse transform method is not directly usable (without numerical approximation).

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- **Box–Muller (1958)** sidesteps this entirely using a clever change of variables based on **polar coordinates**.

Key Idea

Consider two independent standard normal variables (Z_1, Z_2) . Their joint density is rotationally symmetric:

$$f(z_1, z_2) = \frac{1}{2\pi} e^{-(z_1^2 + z_2^2)/2}.$$

Switch to polar coordinates R, Θ where $Z_1 = R \cos \Theta$, $Z_2 = R \sin \Theta$. One can show:

- $R^2 = Z_1^2 + Z_2^2 \sim \text{Exponential}(1/2)$ (equivalently χ_2^2),
- $\Theta \sim \text{Uniform}(0, 2\pi)$, independent of R .

$\Theta \perp R$

$\sim N(0, 1)$

$$f_{Z_1}(z_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z_1^2}{2}}$$
$$f_{Z_2}(z_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z_2^2}{2}}$$

$Z_1 \perp Z_2$

$$f_{Z_1, Z_2}(z_1, z_2) = f_{Z_1}(z_1) \cdot f_{Z_2}(z_2)$$

$\downarrow$ joint PDF of Z_1, Z_2

$$Z_1 \sim N(0,1) \Rightarrow f_{Z_1, Z_2}(z_1, z_2) = \frac{1}{2\pi} \exp\left(-\frac{z_1^2 + z_2^2}{2}\right)$$

$$\begin{aligned} Z_1 &= R \cos \theta \\ Z_2 &= R \sin \theta \end{aligned} \Rightarrow \begin{aligned} &R^2 \sim \text{Exp}\left(\frac{1}{2}\right) \\ &\theta \sim U[0, 2\pi] \\ &R \perp \theta \text{ ind} \end{aligned}$$

$$\Rightarrow (z_1, z_2) \longrightarrow (r, \theta)$$

$$f_{R, \theta}(r, \theta) = f_{Z_1, Z_2}(z_1(r, \theta), z_2(r, \theta)) \cdot |J|$$

$$z_1^2 + z_2^2 = r^2 (\cos^2 \theta + \sin^2 \theta) = r^2$$

$$= \frac{1}{2\pi} \exp\left(-\frac{r^2}{2}\right) \cdot |J|$$

$$\begin{aligned} Z_1 &= R \cos \theta \\ Z_2 &= R \sin \theta \end{aligned}$$

$$|J| = \begin{vmatrix} \frac{\partial z_1}{\partial r} & \frac{\partial z_1}{\partial \theta} \\ \frac{\partial z_2}{\partial r} & \frac{\partial z_2}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta = r$$

$$f_{R, \theta}(r, \theta) = \frac{1}{2\pi} \exp\left(-\frac{r^2}{2}\right) \cdot r$$

Joint
pmf of
 R & θ

$$= r \exp\left(-\frac{r^2}{2}\right) \cdot \frac{1}{2\pi}$$

R.V. X & Y are ind

$\rightarrow$ pdf of θ
 $\rightarrow U(0, 2\pi)$

$$\begin{aligned} \text{when } f_{X,Y}(x,y) &= f_X(x) \cdot f_Y(y) \\ \Leftrightarrow F_{X,Y}(x,y) &= F_X(x) \cdot F_Y(y) \end{aligned}$$

$$= f_R(r) \cdot f_\theta(\theta)$$

$$\Rightarrow R \text{ & } \theta \text{ are independent} \Rightarrow R \sim \text{Exp}\left(-\frac{r^2}{2}\right)$$

$$\Rightarrow R \text{ & } \theta \text{ are independent} \Rightarrow \theta \sim U[0, 2\pi]$$

$$\begin{array}{c} R \rightarrow R^2 \\ R \rightarrow X \end{array} \Rightarrow \frac{dR}{dX}$$

we know $R \sim f_R(R) = \frac{1}{R} \exp(-\frac{R^2}{2})$

what will be the distⁿ of $R^2 \sim \exp(-\frac{1}{2})$

$$X = R^2 \rightarrow \text{this is ??} \Rightarrow R = \sqrt{X}$$

you know distⁿ of R

$$\begin{aligned} f_X(x) &= f_R(R(x)) |J| \rightarrow |J| = \frac{dR}{dX} = \frac{1}{2\sqrt{x}} \\ &= \sqrt{x} \exp\left(-\frac{x}{2}\right) \times \frac{1}{2\sqrt{x}} \\ &= \frac{1}{2} \exp\left(-\frac{x}{2}\right) \end{aligned}$$

$$\begin{aligned} &\rightarrow \text{Exp}(\lambda = \frac{1}{2}) \\ \text{Exp}(\lambda) &\rightarrow f_X(x) = \lambda e^{-\lambda x} \end{aligned}$$

$$\Rightarrow R^2 \sim \text{Exp}\left(\frac{1}{2}\right) \quad \text{ind}$$

$$\Theta \sim U[0, 2\pi]$$

we know
how to
simulate

$R^2 \rightarrow$ inverse transform method

$$\Theta \sim U[0, 2\pi] \rightarrow U \sim U[0, 1]$$

$$\Theta = 2\pi U \sim U[0, 2\pi]$$

$$\Rightarrow \begin{aligned} Z_1 &= R \cos \Theta \\ Z_2 &= R \sin \Theta \end{aligned} \Rightarrow$$

$$\frac{-\log(U)}{1}$$

Alg: 1) Simulate of $R^2 \rightarrow$

$$\text{inverse CDF} \rightarrow R^2 = \frac{-\log(U)}{1}$$

$$\text{Simulate } U_1 \sim U[0, 1] \text{ and set } \underline{R^2 = -2\log(U_1)}$$

$$R = \sqrt{-2\log(U_1)}$$

2) Simulate of Θ

$$U_2 \sim U[0, 1] \Rightarrow \Theta = 2\pi U_2$$

$$(3) \Rightarrow \begin{aligned} Z_1 &= R \cos \Theta \\ Z_2 &= R \sin \Theta \end{aligned}$$

From Polar Back to Cartesian

Since $R^2 \sim \text{Exponential}(1/2)$, by the inverse transform method (Section 3!):

$$R^2 = -2 \ln U_1 \implies R = \sqrt{-2 \ln U_1}, \quad U_1 \sim \text{Unif}(0, 1).$$

And since $\Theta \sim \text{Unif}(0, 2\pi)$:

$$\Theta = 2\pi U_2, \quad U_2 \sim \text{Unif}(0, 1), \text{ independent of } U_1.$$

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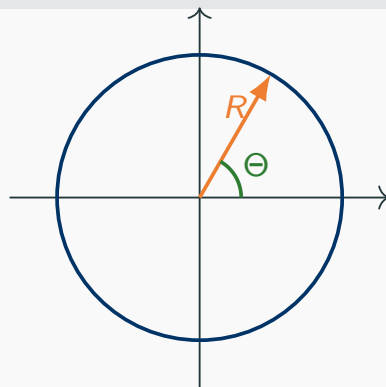
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Box-Muller Transformation

$$Z_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2), \quad Z_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2)$$

gives two **independent** standard Normal variables from two independent uniforms.



Box-Muller: Algorithm

Algorithm: Basic Box-Muller

1. Generate $U_1, U_2 \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$.

2. Compute

$$\underline{Z_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2)}, \quad \underline{Z_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2)}.$$

3. Return Z_1, Z_2 (a free “bonus” second draw at no extra cost!).

$$Z_1 \sim N(0, 1)$$

Task: sample from $N(\mu, \sigma^2)$

$$X \sim N(\mu, \sigma^2)$$

$$\Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$\Rightarrow X = \mu + \sigma Z$$

$$X = \mu + \sigma Z \sim N(\mu, \sigma^2)$$

we know $Z \sim N(0, 1) \rightarrow$ how to
simulate.

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$$Z_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2), \quad Z_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2).$$

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General Normal $N(\mu, \sigma^2)$

$$X = \mu + \sigma Z$$

for Z generated above – a simple location-scale shift.

Practical note

Computing $\cos$, $\sin$, and $\ln$ is relatively expensive. This motivates a trig-free variant next.